

AP MICROECONOMICS · THE UNIVERSAL STUDY GUIDE

Targeting a 3, 4, or 5 — every concept tagged for your tier

Tier badges: [3] foundational · [4] adds for 4+ · [5] adds for 5

Targeting a 4 → study [3] AND [4] items. Targeting a 5 → study all three tiers.

PAGE 1 | MATH OF 3/4/5 · 6 UNITS · MASTER GRAPHS

About this exam. AP Microeconomics studies individual decision-makers — consumers, firms, markets — and how prices coordinate their choices. Unlike Macro (aggregate behavior), Micro emphasizes the analysis of specific markets and the FOUR market structures.

THE MATH OF YOUR TARGET

Target	MCQ (60 q, 66%)	Long FRQ (10 pts, 17%)	Short FRQs (5 pts each, 17%)	Tier
3	~32 / 60 (~53%)	~6 / 10	~5-6 / 10 combined	[3] only
4	~38 / 60 (~63%)	~7-8 / 10	~7-8 / 10	[3] + [4]
5	~46 / 60 (~76%)	~9 / 10	~9 / 10	All

THE 6 UNITS — by exam weight

#	Unit	Weight	Key topics
1	Basic Economic Concepts	5–10%	Scarcity, opportunity cost, PPC, comparative advantage
2	Supply and Demand	20–25%	Equilibrium, shifters, elasticity , consumer/producer surplus
3	Production, Cost, Perfect Competition	22–25%	MR=MC , profit max, short vs long run, cost curves
4	Imperfect Competition	15–22%	Monopoly, monopolistic competition, oligopoly
5	Factor Markets	10–13%	Labor markets, MRP, wage determination
6	Market Failure + Role of Government	8–13%	Externalities, public goods, taxation, income inequality

Units 2, 3, and 4 are 57-72% combined. Supply/demand mechanics, perfect competition, and the four market structures dominate.

THE 5 MASTER GRAPHS — every FRQ uses one [3]/[4]

Graph	Y-axis	X-axis	What it models
Supply & Demand [3]	Price (P)	Quantity (Q)	Market equilibrium, shifters, elasticity
Cost curves (firm) [3]	Cost / Price	Quantity (Q)	ATC, AVC, MC, MR — finding profit-max
Perfect Competition: firm + market [4]	Price	Quantity	Side-by-side: market sets price; firm chooses Q
Monopoly [4]	Price / Cost	Quantity	MR ≠ Demand. MR=MC for Q; charge price from D curve
Labor market [4]	Wage (W)	Quantity of labor (L)	MRP = W in competitive labor market

KEY INSIGHT [5] On every FRQ, label graphs precisely. Always: axes labeled, all curves named, equilibrium marked, areas (CS/PS/DWL) shaded if asked. Unlabeled graphs lose points fast.

PAGE 2 | MCQ MASTERY + MASTER VOCABULARY

MCQ STRATEGY — 60 questions in 70 min [3]/[4]

- [3] **Identify the model first.** Q is about firm behavior? → cost curves. Q is about market efficiency? → consumer/producer surplus. Q is about wages? → labor market.
- [3] **Know the elasticity numbers.** Elastic = >1 ; inelastic = <1 ; unit elastic = 1. Sign of cross-price = relationship type.
- [3] **Profit maximization rule** is universal: $MR = MC$. Differs by market structure only in the MR curve.
- [4] **Direction questions** — "What happens to consumer surplus when...?" Trace through the graph: shift cause → equilibrium change → area calculation.
- [4] **No calculator allowed.** Practice arithmetic — especially elasticity calculations + simple $MR=MC$ tables.

MASTER VOCABULARY — high-frequency Micro terms [3]/[4]

Term	Definition + tier
Opportunity cost [3]	Value of next-best alternative forgone. Every choice has one.
Comparative advantage [3]	Lower opportunity cost. Trade based on this (not absolute advantage).
Marginal analysis [3]	Decision-making by comparing MARGINAL benefit to MARGINAL cost. Continue if $MB > MC$; stop when $MB = MC$.
Elasticity (price) [3]	% change in Q / % change in P. Elastic: $ E > 1$ (luxuries). Inelastic: $ E < 1$ (necessities).
Cross-price elasticity [4]	Positive = substitutes. Negative = complements. Magnitude = strength of relationship.
Income elasticity [4]	Positive = normal good. Negative = inferior good. >1 = luxury.
Consumer surplus [3]	Area below demand curve, above price. The benefit consumers get above what they paid.
Producer surplus [3]	Area above supply curve, below price. The benefit producers get above their costs.
Total surplus / efficiency [4]	CS + PS. Maximized at competitive equilibrium. Allocative efficiency: $P=MC$.
Deadweight loss (DWL) [3]	Lost surplus due to inefficiency (taxes, monopolies, price ceilings/floors).
Marginal cost (MC) [3]	Cost of producing one more unit. $\Delta TC / \Delta Q$. MC curve intersects ATC and AVC at their minimums.
ATC, AVC, AFC [3]	Average total/variable/fixed cost. $ATC = TC/Q$. AFC always declining (FC spread over more units).
Profit-maximizing rule [3]	Produce Q where $MR = MC$. Universal across all market structures.
Shut-down rule [4]	In SHORT RUN: produce if $P \geq AVC$. Shut down if $P < AVC$. Lose less by shutting down (only fixed costs).
Long-run zero-profit condition [4]	In perfect competition, $P = ATC$ at the minimum of ATC. Firms enter/exit until normal profit only.
Economies of scale [4]	ATC falls as Q rises (LR). Common in capital-intensive industries.
Diminishing marginal returns [3]	Adding more of one input (with others fixed) yields smaller additional output. Causes MC to rise.
Price ceiling vs floor [3]	Ceiling = max price (rent control); below equilib creates shortage. Floor = min price (min wage); above creates surplus.
Tax incidence [4]	Who really pays? Burden falls more on the inelastic side. Inelastic demand → consumers pay more.
Marshallian Cross [4]	Standard supply-demand graph (Alfred Marshall).
Marginal Revenue Product (MRP) [4]	$MR \times MP_L$. Demand curve for labor.
Externality [3]	Third-party effect not in market price. Negative (pollution); positive (vaccines).
Pigouvian tax [4]	Tax equal to negative externality cost. Forces internalization. Optimal = marginal external cost.
Public good [3]	Non-excludable + non-rivalrous. Free-rider problem → underprovision by markets.
Common resource [4]	Rivalrous but non-excludable. Tragedy of the commons.
Lorenz curve / Gini coefficient [4]	Measure of income inequality. Gini 0 = perfect equality; 1 = max inequality.

UNITS 1–2 | BASIC CONCEPTS + SUPPLY AND DEMAND

Unit 1 — Basic Concepts [3]/[4]

- **Scarcity** = unlimited wants vs limited resources. The fundamental economic problem.
- **Opportunity cost [3]** = value of the next-best alternative forgone. Slope of PPC = OC of switching production.
- **PPC [3]** — bowed out (concave) = increasing OC. Linear = constant OC. Inside = inefficient. Outside = unattainable.
- **Comparative advantage [3]** — lower OC. Trade improves both parties when each specializes in their CA. **Terms of trade must lie BETWEEN the two parties' OCs.**

Unit 2 — Supply, Demand, Equilibrium [3]

DEMAND — downward sloping. Inverse relationship between P and Q demanded.

Shifters of D: T-R-I-B-E

T - Tastes / preferences

R - Related goods (substitutes $\uparrow P \rightarrow D$ right; complements $\uparrow P \rightarrow D$ left)

I - Income (normal: $\uparrow I \rightarrow D$ right; inferior: $\uparrow I \rightarrow D$ left)

B - Buyers (more buyers $\rightarrow D$ right)

E - Expectations (expect $\uparrow P \rightarrow D$ right NOW)

SUPPLY — upward sloping. Direct relationship between P and Q supplied.

Shifters of S: R-O-T-T-E-N

R - Resource costs ($\uparrow$ input prices $\rightarrow S$ left)

O - Other goods firm could produce

T - Technology

T - Taxes/subsidies (taxes $\rightarrow S$ left; subsidies $\rightarrow S$ right)

E - Expectations

N - Number of sellers

KEY INSIGHT [4] Critical distinction: Movement ALONG the curve (price changes) vs SHIFT of the curve (anything else). Don't say "demand decreased because price went up" — that's a movement, not a shift.

Elasticity [3]/[4]

PRICE ELASTICITY OF DEMAND (PED): $|\% \Delta Q_d / \% \Delta P|$

>1 = ELASTIC (luxuries, lots of substitutes)

<1 = INELASTIC (necessities, addictive goods)

$=1$ = Unit elastic

Total revenue test: when D is elastic, $\uparrow P \rightarrow \downarrow TR$; when inelastic, $\uparrow P \rightarrow \uparrow TR$

CROSS-PRICE ELASTICITY: $\% \Delta Q_a / \% \Delta P_b$

Positive $\rightarrow$ SUBSTITUTES (Pepsi vs Coke)

Negative $\rightarrow$ COMPLEMENTS (peanut butter + jelly)

INCOME ELASTICITY: $\% \Delta Q / \% \Delta I$

Positive $\rightarrow$ Normal good

Positive >1 $\rightarrow$ Luxury

Negative $\rightarrow$ Inferior good

Consumer + Producer Surplus, DWL [3]/[4]

- **Consumer surplus** = triangle area under D curve, above price. **Maximum when market is competitive equilibrium.**
- **Producer surplus** = triangle area above S curve, below price. **Same maximum condition.**
- **Total surplus = CS + PS.** Maximized at competitive equilibrium $\rightarrow$ ALLOCATIVE EFFICIENCY ($P=MC$).
- **Deadweight loss (DWL)** = lost total surplus from market inefficiency (price controls, taxes, monopolies). Triangle between curves at non-equilibrium quantity.

Government in Markets [3]/[4]

- **Price ceiling [3]** (max price, e.g., rent control). If BELOW equilibrium $\rightarrow$ SHORTAGE. Creates DWL.
- **Price floor [3]** (min price, e.g., minimum wage). If ABOVE equilibrium $\rightarrow$ SURPLUS. Creates DWL.
- **Per-unit tax [4]** — vertical wedge between price buyers pay and price sellers receive. Burden split based on elasticities. Reduces Q below efficient. Creates DWL.
- **Subsidy [4]** — opposite of tax. Increases Q above efficient. Also creates DWL (overproduction).

UNIT 3 | PRODUCTION, COST, PERFECT COMPETITION

Production Function + Returns [3]/[4]

- **Marginal product (MP)** = $\Delta \text{Total Output} / \Delta \text{Labor}$. Eventually FALLS due to diminishing returns.
- **Diminishing marginal returns [3]** — adding more of one input (labor) to fixed others (capital) eventually yields smaller additional output. **This causes MC to rise.**
- **Returns to scale (LR) [4]** — increasing returns (output $\uparrow$ proportionally more than inputs); constant returns; decreasing returns. Different from diminishing marginal returns.

Cost Curves — the family [3]/[4]

Total Cost (TC) = TFC + TVC
 Avg Fixed Cost = TFC/Q (always declining as Q rises)
 Avg Variable Cost = TVC/Q (U-shaped)
 Avg Total Cost = $\text{TC}/Q = \text{AFC} + \text{AVC}$
 Marginal Cost = $\Delta \text{TC}/\Delta Q = \Delta \text{TVC}/\Delta Q$

KEY GEOMETRY:

MC intersects AVC at AVC's MINIMUM
 MC intersects ATC at ATC's MINIMUM
 Below the minimums, $\text{MC} < \text{AVC}$ (or ATC), pulling them DOWN
 Above the minimums, $\text{MC} > \text{AVC}$ (or ATC), pulling them UP

Profit Maximization [3]

UNIVERSAL RULE: produce Q where $\text{MR} = \text{MC}$
 (the last unit's revenue covers its cost)

Then: Profit = $(P - \text{ATC}) \times Q$
 = $\text{TR} - \text{TC}$

In short run, even unprofitable firms may operate (cover variable costs).
 Shut-down rule: Operate if $P \geq \text{AVC}$. Shut down if $P < \text{AVC}$.
 (When $P < \text{AVC}$, you lose less by closing than by producing.)

Perfect Competition — characteristics [3]/[4]

- **Many buyers and sellers** — no one has market power.
- **Identical (homogeneous) products** — wheat, copper, eggs.
- **Free entry and exit** in long run.
- **Perfect information** + perfect mobility of factors.
- **Firm is a price taker** — accepts market price; demand for individual firm = horizontal line at market P.

Perfect Competition — Short vs Long Run [4]

IN PERFECT COMPETITION: $P = \text{MR} = \text{AR}$ (firm faces horizontal demand)

SHORT RUN: firms can earn profit, loss, or break-even

Profit: $P > \text{ATC}$ at Q^* (firm earns economic profit)

Loss: $\text{AVC} \leq P < \text{ATC}$ at Q^* (firm operates but loses; loses less than shutting down)

Shutdown: $P < \text{AVC}$ (lose only fixed costs; close)

LONG RUN: free entry/exit drives P to ATC minimum

Firms enter when there are profits $\rightarrow S$ right $\rightarrow P$ falls

Firms exit when there are losses $\rightarrow S$ left $\rightarrow P$ rises

Equilibrium: $P = \text{MC} = \text{ATC}_{\min}$ (zero economic profit; "normal" return)

PERFECT COMPETITION = ALLOCATIVE EFFICIENT ($P = \text{MC}$) AND PRODUCTIVELY EFFICIENT ($P = \text{ATC}_{\min}$)

KEY INSIGHT [5] Productive efficiency vs allocative efficiency. **Productive** = producing at lowest cost ($P = \text{ATC}_{\min}$). **Allocative** = producing the right Q ($P = \text{MC}$). Perfect competition achieves BOTH in the long run. Most other market structures fail at one or both.

UNIT 4 | IMPERFECT COMPETITION · the 4 market structures

[4] AP Micro's signature topic. Master the comparison among the four market structures — this is what shows up on FRQs constantly.

THE 4 MARKET STRUCTURES — side-by-side comparison [4]/[5]

Feature	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
# of firms	Many	Many	Few	One
Product	Identical	Differentiated	Either	Unique (no substitutes)
Entry barriers	None	None / low	High	Very high / blocked
Demand for firm	Horizontal	Downward-sloping (slight)	Kinked / strategic	Downward-sloping (steep)
MR vs Demand	$MR = D = P$	$MR < D$ (below D)	$MR < D$	$MR < D$ (significantly below)
LR profit	Zero (entry/exit)	Zero (similar reason)	Possible profit	Possible long-run profit (barriers)
Allocative effic.?	Yes ($P=MC$)	No ($P>MC$)	No	No ($P>>MC$)
Productive effic.?	Yes ($P=ATC_{min}$)	No (excess capacity)	Variable	No
Examples	Wheat, copper	Restaurants, clothing	Airlines, autos, telecoms	Public utilities, patents

MONOPOLY — the deepest dive [4]/[5]

Monopoly faces the **WHOLE** market demand (downward-sloping).
Therefore $MR < D$ (below it on the graph) — to sell more, must lower P on ALL units.

Profit-max: produce Q where $MR = MC$
Charge price P from the **DEMAND** curve at that Q (NOT from MR curve)

Result: $P > MC \rightarrow$ **ALLOCATIVE INEFFICIENCY** $\rightarrow$ **DEADWEIGHT LOSS**

Compare to PC: PC produces more Q at lower $P \rightarrow$ larger CS , no DWL

- [4] **Sources of monopoly**: natural (economies of scale, e.g., utilities); legal (patents); resource ownership; government license.
- [4] **Price discrimination** [5]: charging different prices to different consumers. Requires market power + ability to segment + prevent resale. Increases monopoly profits, can reduce DWL .

MONOPOLISTIC COMPETITION [4]

Many sellers + differentiated products + free entry/exit. Examples: restaurants, clothing brands.

- **Short run** — like monopoly (price $> MC$, can earn profit if differentiation is valuable).
- **Long run** — free entry erodes profit. $P = ATC$, but $P > MC$ and $P > ATC_{min}$. **Excess capacity** + allocative inefficiency persist due to differentiation.

OLIGOPOLY — strategic interaction [4]/[5]

- **Few firms, mutual interdependence**. Each firm's decision affects rivals. Examples: airlines, smartphones.
- [4] **Game theory** — Prisoner's Dilemma is the classic model. Both firms have dominant strategy to defect (compete), but mutual cooperation would be better.
- [5] **Nash equilibrium** — strategy combo where no firm can improve by changing alone.
- [5] **Collusion vs cartels** — explicit collusion illegal in US; tacit collusion harder to prove. Cartels (OPEC) often unstable due to incentive to cheat.

UNIT 5 | FACTOR MARKETS · labor, land, capital

Factor markets work like product markets, but in reverse: firms DEMAND labor; workers SUPPLY labor. Equilibrium wage equates the two.

Marginal Revenue Product (MRP) — the labor demand curve [4]

$MRP_L = \text{Marginal Product of Labor (MP}_L) \times \text{Marginal Revenue (MR)}$

In perfect competition (output market): $MR = P$, so $MRP_L = MP_L \times P$

In monopoly: $MR < P$, so MRP_L is lower

MRP_L IS THE FIRM'S DEMAND CURVE FOR LABOR (downward-sloping due to diminishing MP).

HIRE workers up to where $MRP_L = W$ (wage)

Hire one more worker if $MRP_L > W$ (worker adds more value than cost)

Stop hiring when $MRP_L = W$ (last worker breaks even)

Labor Market Structures [4]/[5]

Structure	Outcome
Perfectly competitive labor market [4]	Many firms hiring, many workers. W set by market; firm is wage taker. $W = MRP$ for last worker hired.
Monopsony [5]	ONE buyer of labor (e.g., dominant employer in town). To hire more, must raise W for ALL workers. Hires LESS labor at LOWER wage than competitive market.
Union (monopoly seller of labor) [5]	Workers organize collectively. Can raise wage above competitive level — but creates unemployment (excess workers at union wage).

Factor Market Shifters [4]

- **Demand for labor shifts** when: product price changes ($\uparrow P \rightarrow \uparrow MRP$); productivity changes (tech); price of complementary factors changes; demand for output changes.
- **Supply of labor shifts** when: working-age population; immigration; preferences for work vs leisure; non-wage compensation; alternative wages elsewhere.

Income Distribution + Inequality [4]/[5]

- **Lorenz curve** [4]: cumulative income share vs cumulative population share. Diagonal = perfect equality. Bowed = inequality.
- **Gini coefficient** [4]: 0 = perfect equality; 1 = max inequality. Most countries: 0.25-0.55.
- **Sources of inequality** [5]: differences in human capital, discrimination, inheritance, market vs labor luck, demand for skills.

UNIT 6 | MARKET FAILURE + ROLE OF GOVERNMENT

Externalities [3]/[4]

Externality = effect on a third party not in the price signal. Markets fail to internalize these → wrong quantity produced.

NEGATIVE externality (e.g., pollution):

Marginal Social Cost (MSC) > Marginal Private Cost (MPC)

Market produces TOO MUCH (private equilibrium > socially optimal)

DWL = triangle between MSC and demand at extra units produced

SOLUTION: Pigouvian TAX equal to marginal external cost

POSITIVE externality (e.g., education, vaccines):

Marginal Social Benefit (MSB) > Marginal Private Benefit (MPB)

Market produces TOO LITTLE

SOLUTION: SUBSIDY equal to marginal external benefit

KEY INSIGHT [5] Coase theorem — If property rights are well-defined and transaction costs are zero, parties can NEGOTIATE efficient outcome regardless of who has the rights. **Reality** check: transaction costs often high → government intervention needed.

Public Goods [3]/[4]

- **Public goods [3]** = non-excludable + non-rivalrous. National defense, lighthouses, public TV. **Free-rider problem** → markets underprovide → government provision.
- **Common resources [4]** = rivalrous but non-excludable (fish, grazing land). **Tragedy of the commons** = overuse without intervention.
- **Private goods [3]** = excludable + rivalrous (most goods).
- **Club goods [4]** = excludable but non-rivalrous (cable TV, gym memberships).

The 4 Types of Goods [4]

Good Type	Excludable?	Rivalrous?
Private	YES	YES (food, clothing)
Club	YES	NO (cable, gym)
Common	NO (tragedy of commons)	YES (fish, grazing)
Public	NO (free riders)	NO (defense, fireworks)

Taxation + Income Redistribution [4]

- **Progressive tax [3]** — higher rate as income rises (US income tax).
- **Regressive tax [3]** — lower effective rate as income rises (sales tax — poor spend higher % of income).
- **Proportional / flat tax [3]** — same rate for all.
- **Tax incidence [3]** — who actually bears the burden depends on elasticities. Inelastic side bears more burden.

PAGE 8 | FRQ STRATEGY · graphing + chains of reasoning

AP Micro FRQ Structure [3]

1 long FRQ (10 pts, 25 min) + 2 short FRQs (5 pts each, 12 min each). Total: 60 min for 20 raw points.

UNIVERSAL FRQ DISCIPLINE [3]/[4]

- [3] If asked to "draw" a graph — **DRAW IT**. No graph = no points on that part. Label axes precisely.
- [3] Label **EVERY** curve (S, D, MR, MC, ATC, AVC, MRP_L, etc.) and equilibrium points.
- [3] Specify **direction of shifts** with arrows. "Curve shifts to the right" or arrow on graph.
- [4] State **direction of price/quantity changes**: "P increases" not just "P changes."
- [5] Explain **WHY** by linking to mechanism: "This shifts demand right because consumer income increased and the good is normal."

CHAIN OF REASONING — universal Micro template [4]/[5]

Stimulus → Affected curve → Direction of shift → New equilibrium →
Consequence for surplus / efficiency / quantity / price

Example: "Government imposes a \$2/unit tax on producers."

1. Tax acts on the supply side
2. Supply shifts **LEFT** by \$2 vertically
3. New equilibrium: higher P paid by consumers, lower P received by producers
4. Quantity transacted falls
5. Consumer surplus falls; producer surplus falls
6. Government collects tax revenue (rectangle)
7. **DEADWEIGHT LOSS** = lost trades (triangle)

THE 7 GRAPHS YOU MIGHT DRAW [4]

- **Supply & Demand market** — most common. Show shifts, equilibrium changes, CS/PS/DWL.
- **Perfect competition: market + firm** — side-by-side. Market sets P; firm chooses Q where MR=MC.
- **Monopoly graph** — D, MR, MC, ATC. MR=MC for Q; price from D curve at that Q.
- **Monopolistic competition long run** — like monopoly but P=ATC at chosen Q (zero economic profit).
- **Externality (negative)** — MSC above MPC. Show market vs socially optimal Q + DWL triangle.
- **Labor market** — competitive: D=MRP, S=labor supply, equilibrium W and L.
- **Monopsony [5]** — labor supply curve is upward; MFC (marginal factor cost) above it. Hire where MRP=MFC; pay wage from supply curve.

PAGE 9 | THE 3/4/5 SCORE LADDER · tier-specific test-day

THE SCORE LADDER

3-scorer writes...	4-scorer writes...	5-scorer writes...
"S shifts left, P goes up."	"The tax causes Supply to shift left vertically by the tax amount, raising the price consumers pay and lowering the price producers receive."	"The tax shifts S left by the tax amount, splitting price into a higher consumer price and lower producer price. Tax incidence depends on elasticities — the more inelastic side bears more of the burden. DWL = triangle of lost trades at quantities where MB exceeded MC pre-tax but not post-tax. The tax revenue (rectangle) transfers surplus from consumers + producers to government but net surplus falls ."
"Monopoly is bad."	"Monopoly produces less Q at higher P than perfect competition, creating DWL."	"Monopoly's profit-max condition ($MR=MC$ where $MR < D$) yields lower Q and higher P than perfect competition's equilibrium ($P=MC$). Allocative inefficiency ($P>MC$) creates DWL. Productive inefficiency if monopolist doesn't operate at ATC_{min} . Welfare consequences: higher CS loss > monopoly profit gain → net welfare loss. Counterargument [5]: natural monopoly may be more efficient than competition due to economies of scale."
States $MR=MC$	Identifies $MR=MC$ AND interprets graphically	Distinguishes which market structure's MR curve to use AND explains why the rule applies universally despite different MR curves
Draws supply and demand	Draws + labels axes + curves + equilibrium with letters	Adds: shaded surplus areas + DWL + arrows showing shifts + clear labels for ORIGINAL vs NEW equilibrium

TIER-SPECIFIC TEST-DAY CHECKLISTS

For a 3 — bag ALL of these:

- ✓ MCQ: 32+ correct. Don't leave anything blank.
- ✓ Each FRQ: draw the graph when asked. Label axes.
- ✓ Memorize $MR=MC$ profit-max rule. Know the 4 market structures by name.
- ✓ Identify direction of shifts from descriptions.

For a 4 — add these on top:

- ✓ MCQ: 38+ correct. Drill elasticity calculations + cost curve relationships.
- ✓ Each FRQ: label ALL curves, equilibria, and shaded surplus areas correctly.
- ✓ Interpret graph mechanically: shift cause → equilibrium change → P/Q effects.
- ✓ Distinguish allocative vs productive efficiency.
- ✓ Master the Market Structure Comparison Table (Page 5).

For a 5 — add these on top:

- ✓ MCQ: 46+ correct.
- ✓ Every FRQ: full chain of reasoning explaining WHY.
- ✓ Connect surplus/efficiency analysis to welfare conclusions.
- ✓ Use technical terms: "allocative inefficiency," "DWL," " $P > MC$," "long-run zero economic profit."
- ✓ For factor markets: distinguish competitive vs monopsony outcomes.
- ✓ For game theory: identify dominant strategies and Nash equilibrium.

THE ONE THING THAT MATTERS MOST [3][4][5]: GRAPH + EXPLAIN. AP Micro grades graphs heavily. Every FRQ has graphical components. Always: (1) draw the graph cleanly with labels, (2) explain in words what the graph shows, (3) state direction of changes. A correct graph with no explanation earns partial credit; an explanation without a graph often earns zero.

If you do nothing else: master the 4 market structures (Page 5), draw and label every graph, and apply $MR=MC$ universally. That's how you hit your target.

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